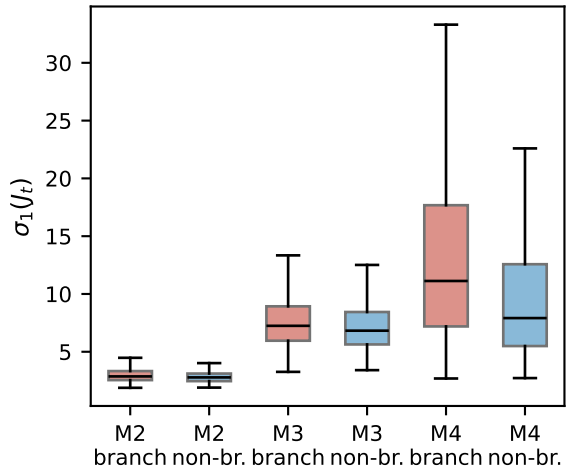


(a) expansion at branch vs non-branch steps



(b) branch prediction, model vs pause nulls

